

Deliverable 2 (D2)

Campus(NC/OC)	OC
Project ID	
Project Title	StreetLight - “No civic hazard should remain in dark”
Team Lead Roll Num	BITF22M015
Supervisor	Sir Adeel Nisar

Capstone Project Deliverable-2

STREETLIGHT

“No civic hazard should remain in dark”

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EXECUTIVE SUMMARY

Introduction:

StreetLight is Pakistan's first end-to-end digital platform for civic infrastructure management, connecting citizens with municipal authorities to resolve everyday problems. The system uses mobile reporting, automated analysis, community validation, and secure storage to ensure civic issues from garbage to potholes are properly reported, verified, assigned, resolved, and documented, creating transparency for citizens and providing municipalities with the reliable data they need for effective urban planning.

Problem Definition:

In Pakistan, citizens face daily civic issues like garbage accumulation, potholes, broken roads, and faulty streetlights that affect their safety and quality of life, yet there is no effective platform to report these problems. Current systems rely on phone calls, scattered emails, or outdated methods that are slow and unaccountable, leaving citizens unaware of which department handles their issue or if action is being taken. Meanwhile, municipal departments lack proper data to identify recurring problems or predict risks, forcing them into reactive crisis management rather than proactive planning. This creates a broken system that fails both citizens and authorities.

Solution Description:

StreetLight is a digital platform that connects citizens with municipal departments to fix civic problems faster and more transparently.

- **For Citizens:** Report issues like garbage or potholes by taking a photo. The app captures GPS location automatically and creates a tracking ID. Track complaint progress in real-time from submission to resolution.
- **Community Verification:** Nearby residents vote to confirm if reported problems are real. This prevents fake complaints and ensures only genuine issues reach authorities. Verified reports are automatically sent to the correct department.
- **For Municipal Departments:** View all verified complaints with photos, location, and severity details on a web dashboard. Assign tasks to teams and update status at each step. Upload before-and-after photos as proof of work completion.
- **Smart Features:** AI identifies issue type and urgency for better prioritization. Automatic duplicate detection combines similar reports to save effort. Blockchain stores verified records permanently to prevent tampering and ensure accountability.
- **Data & Planning:** Analytics dashboards show problem hotspots and department performance. Historical data helps cities plan preventive repairs instead of only reacting to emergencies.

Work Plan Overview

Phase	Deliverables	Activities
Phase 1: Foundation	Core Reporting System	- Build mobile app interface - Implement photo upload and GPS capture - Create user registration and login - Develop complaint submission form - Generate unique tracking IDs - Set up database and backend APIs
Phase 2: Verification & Routing	Community Validation & Municipal Dashboard	- Design community voting system - Build verification logic for nearby residents - Create web dashboard layout - Implement complaint viewing and filtering - Add task assignment features - Develop automatic department routing - Set up notification system
Phase 3: Intelligence & Security	AI Integration & Blockchain	- Train AI model for issue classification - Implement severity detection algorithm - Build duplicate complaint detector - Integrate blockchain for record storage - Add proof-of-completion requirements - Develop accountability tracking
Phase 4: Analytics & Optimization	Data Insights & Performance Tools	- Create analytics dashboards with charts - Build interactive maps for hotspot detection - Implement performance metrics tracking - Add data export functionality - Conduct system testing and bug fixes - Deploy to cloud platform - Prepare user documentation

DESIGN DETAILS

Use Case Model

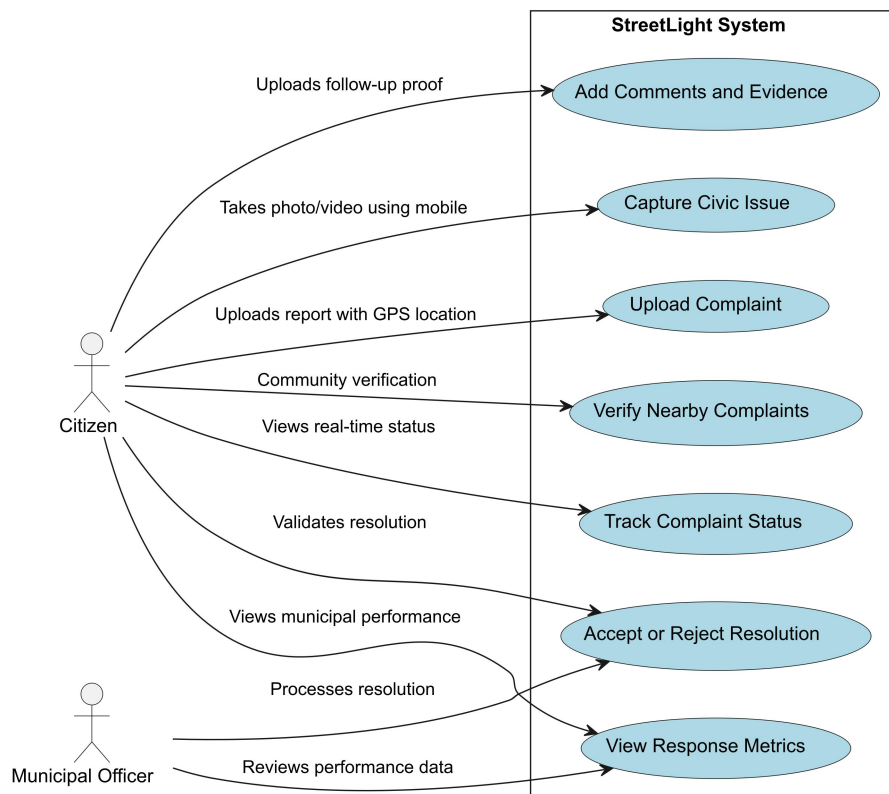
Use Case # 1 (CITIZEN USE CASES)

Actors:

- Citizen (complaint reporter)
- Mobile Application (interface)
- System (backend and notifications)

Description:

Citizens use the mobile app to report civic problems by taking photos, which automatically captures their GPS location. They can verify complaints from nearby residents to confirm issues are real, track their own complaint's progress through different stages, add follow-up comments or photos if needed, and accept or reject the resolution when officers upload proof of completed work. Citizens can also view statistics showing how fast their municipality responds to different problem types, helping them understand government performance.



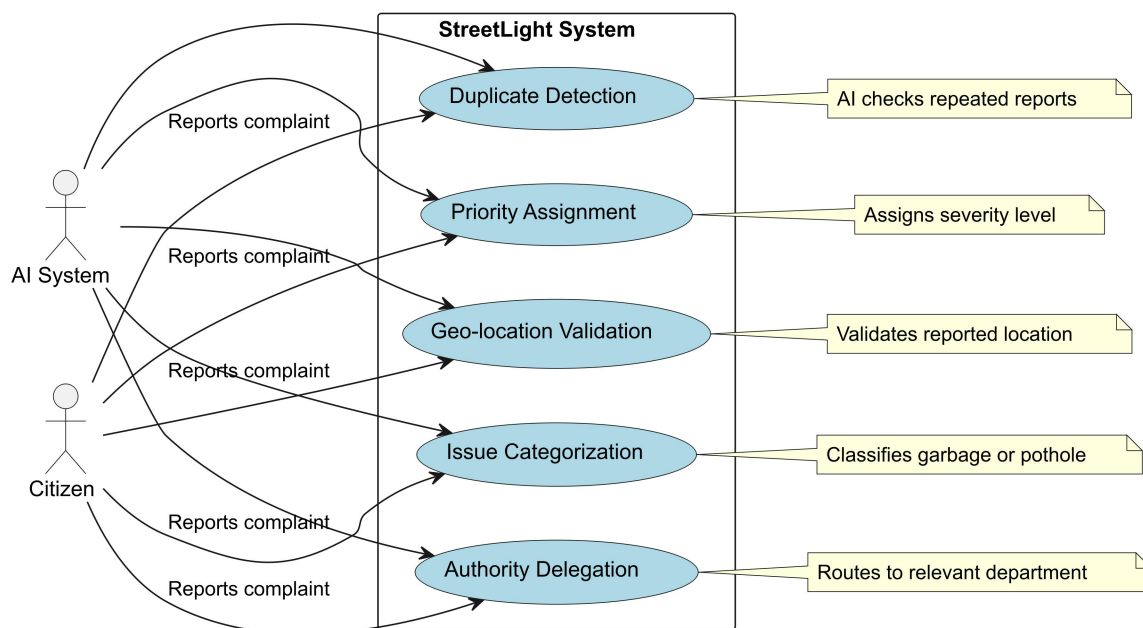
Use Case # 2 (AI & AUTOMATED PROCESSING)

Actors:

- AI Classification Engine (automated system)
- GPS/Maps Service (location provider)
- System Routing Logic (automated dispatcher)

Description:

This group handles all automated tasks that happen without human involvement. When a citizen submits a complaint, AI immediately checks if the same problem was already reported nearby (duplicate detection), figures out what type of issue it is by looking at the photo (categorization), determines how urgent it is (priority assignment), confirms the GPS location is accurate (geo-validation), and automatically sends it to the correct municipal department (authority delegation). This automation saves time and ensures complaints reach the right people instantly.



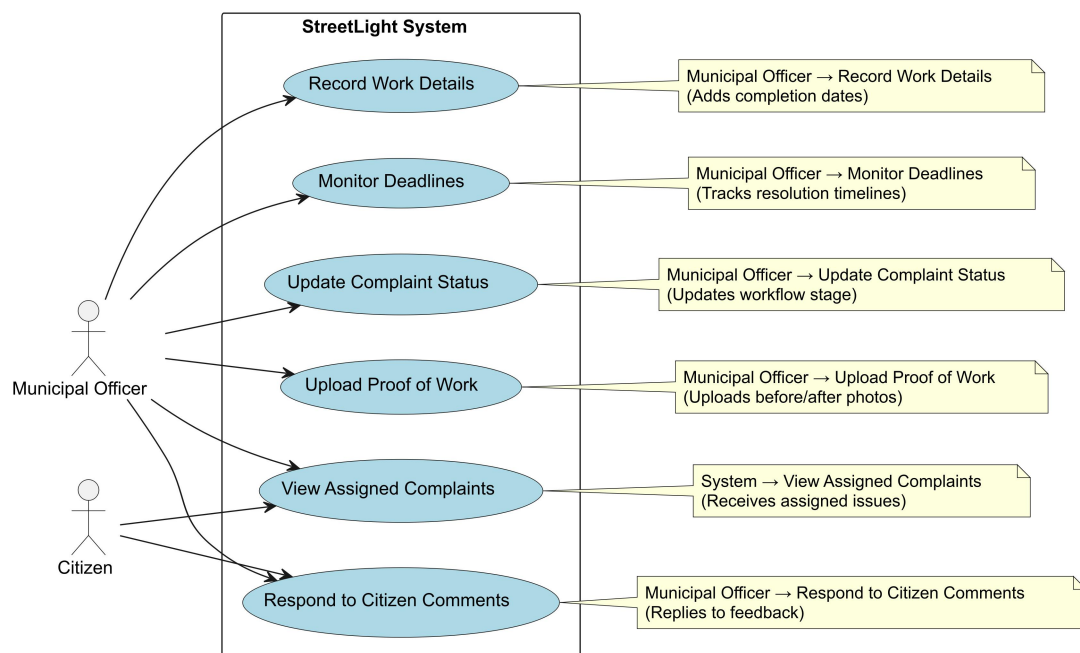
Use Case # 3 (MUNICIPAL OFFICER OPERATIONS)

Actors:

- Municipal Officer (field worker)
- System (task management and notification system)
- Citizen (complaint owner who receives updates)

Description:

These use cases cover everything municipal officers do to fix reported problems. Officers view their assigned tasks, check deadlines to prioritize urgent work, update the complaint status as they progress through repair stages, upload before-and-after photos as proof they completed the work, record details about what was fixed and how long it took, and reply to any questions or comments citizens post on their complaints. All action officers take automatically notifies the citizen who reported the issue, creating transparency and accountability.



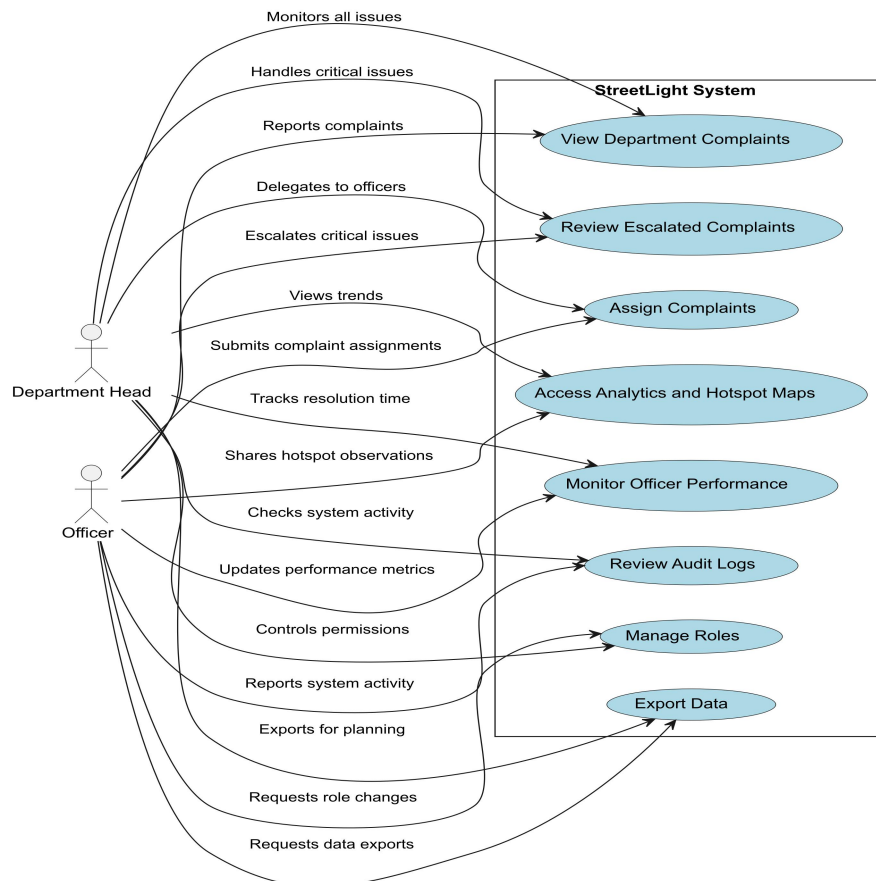
Use Case # 4 (DEPARTMENT HEAD MANAGEMENT)

Actors:

- Department Head (supervisor/manager)
- System (analytics and data platform)
- Municipal Officers (team members)

Description:

Department heads use these functions to oversee their entire team's work and make data-driven decisions. They view all complaints assigned to their department, review escalated cases where deadlines were missed or citizens disputed resolutions, assign new complaints to available officers based on workload and location, access interactive maps showing problem hotspots across the city, monitor how quickly each officer completes tasks, check audit logs to see who did what and when, manage user roles and permissions, and export data for budget planning and performance reports.



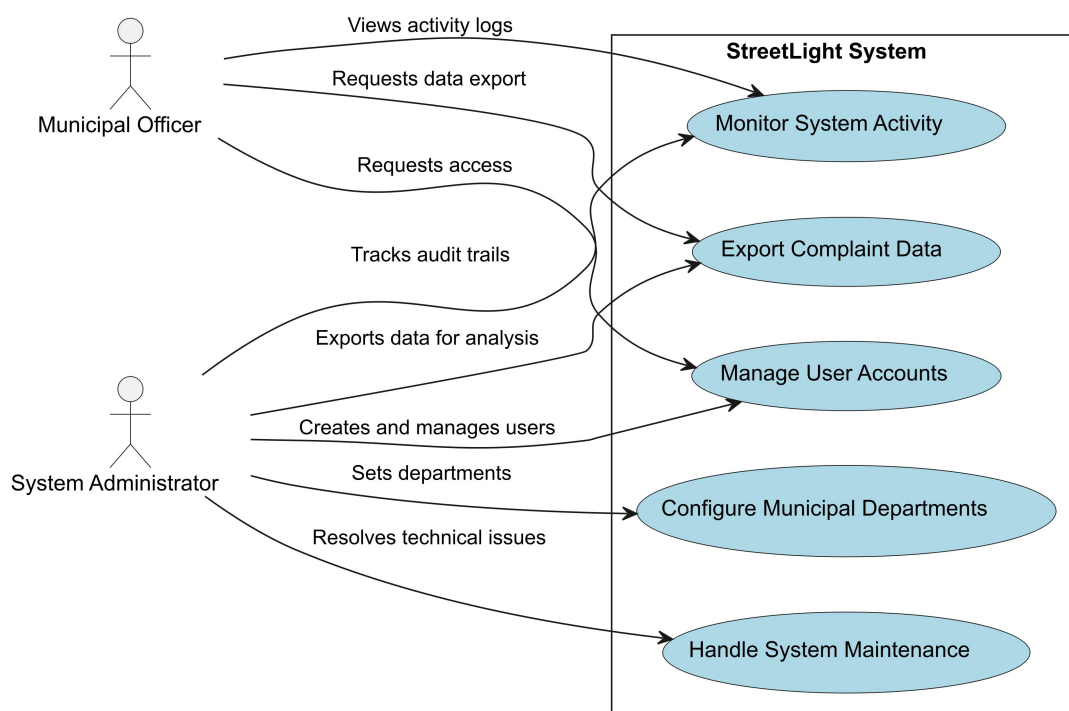
Use Case # 5 (SYSTEM ADMINISTRATION)

Actors:

- System Administrator (technical staff)
- System (platform configuration and monitoring tools)
- All Users (citizens, officers, managers affected by system settings)

Description:

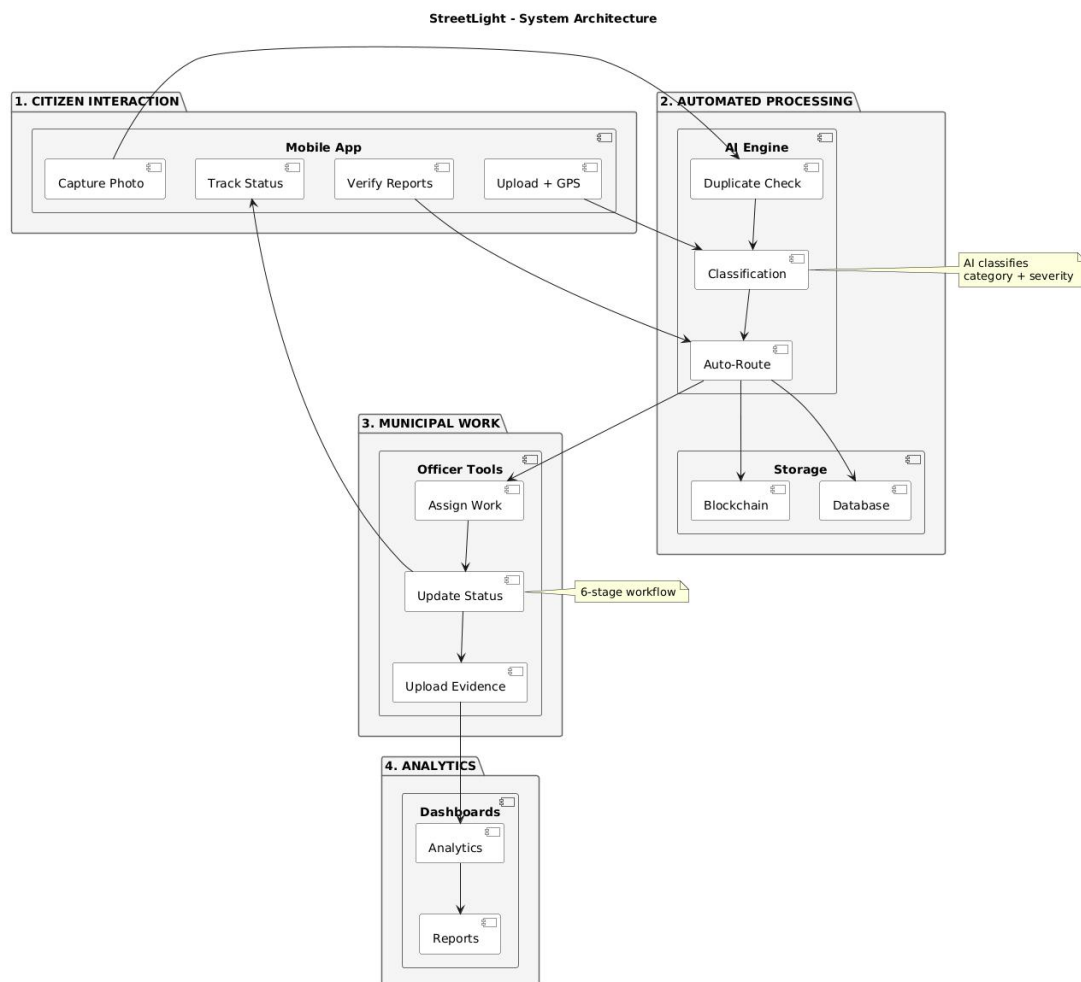
System administrators keep the entire platform running smoothly and properly configured. They monitor all system activity to catch errors or suspicious behavior, export complaint data for external analysis or backups, create and manage user accounts for citizens and government staff, configure municipal departments by setting their names and responsible issue categories, and handle technical maintenance like software updates, database backups, and security patches. These behind-the-scenes tasks ensure the platform stays secure, accurate, and available for everyone to use.



StreetLight - System Use Case

StreetLight operates through four interconnected layers. Citizens use the mobile app to capture photos, upload complaints with GPS, verify nearby reports, and track their complaint status. The automated processing layer runs AI to check for duplicate reports, classify the issue type and severity, and automatically route verified complaints to the correct department while storing records in both the database and blockchain for security.

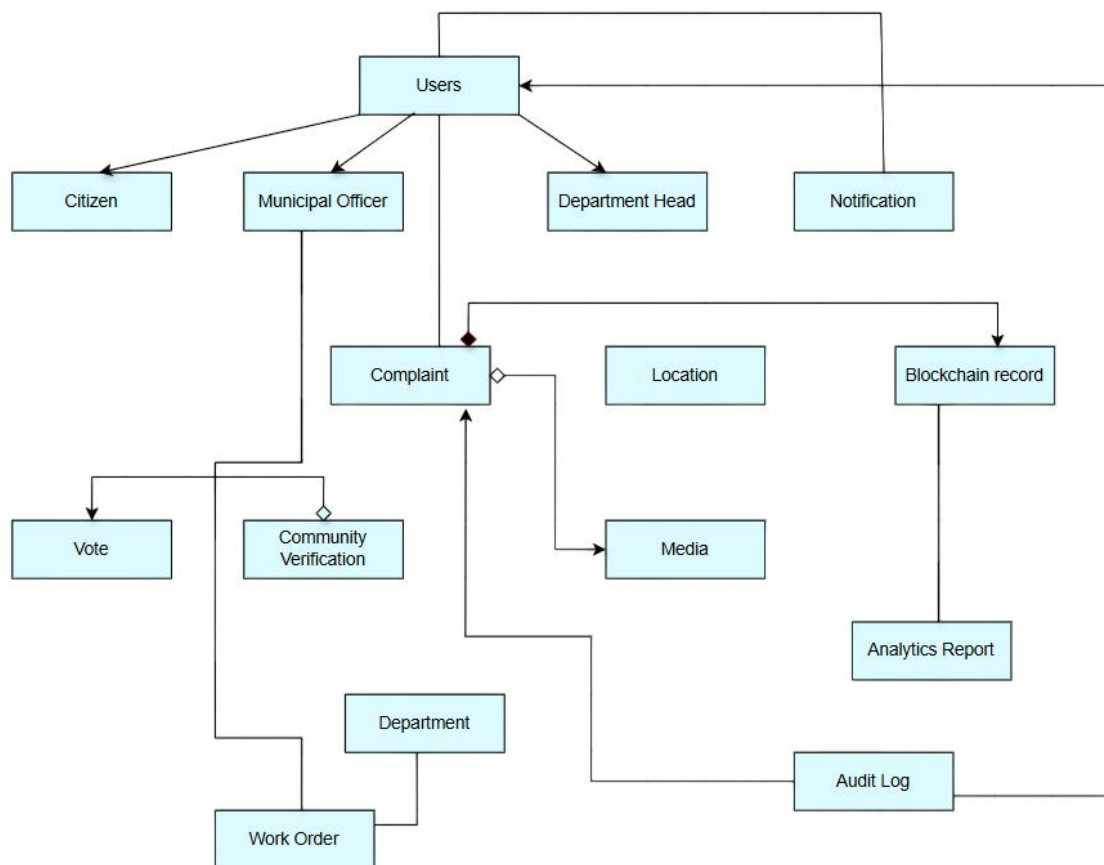
Municipal officers then use their dashboard tools to view assigned work, update the complaint status through a 6-stage workflow, and upload before-and-after evidence photos. Finally, all resolved complaint data feeds into the analytics layer where department head's view dashboards with heatmaps and performance charts, then export reports for urban planning. Data flows smoothly from citizen submission through AI processing, municipal resolution, and analytics insights, creating a complete accountability cycle.



Domain Model/Entity Model

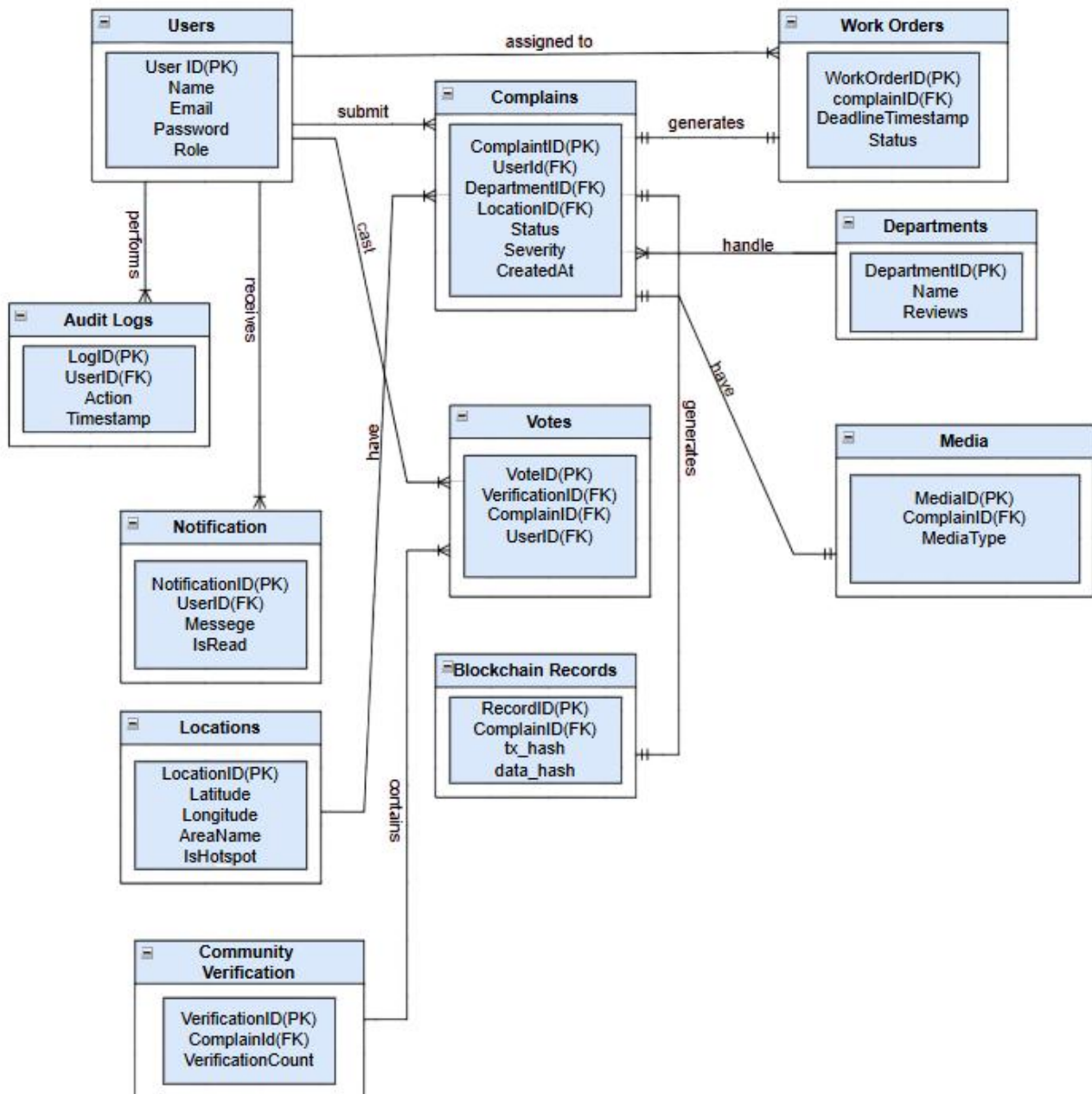
The domain model represents the core entities and their relationships in the StreetLight system. At the center is the **Users** entity, which encompasses three specialized user types:

Citizen (who reports and verifies complaints), **Municipal Officer** (who resolves assigned work), and **Department Head** (who manages teams and reviews analytics). The **Complaint** entity serves as the primary data object, connected to **Location** (GPS coordinates of the issue) and **Media** (photos/videos as evidence). Each complaint goes through **Community Verification** where nearby **Citizens** submit **Votes** to validate authenticity. Verified complaints are assigned to a **Department**, which creates **Work Orders** for officers to execute. Upon resolution, complaints generate **Blockchain Records** for immutability, **Analytics Reports** for performance insights, and **Audit Logs** for accountability tracking. **Notifications** keep all users informed of status changes throughout the workflow.



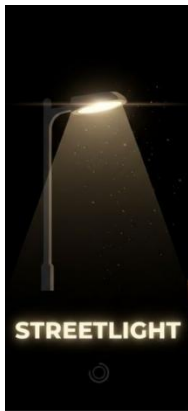
ER Model:

The system has Users who are either Citizens (report problems), Municipal Officers (fix problems), or Department Heads (manage teams). Complaints contain the issue details with Location (GPS) and Media (photos). Citizens cast Votes for Community Verification to confirm complaints are real. Verified complaints go to Departments which create Work Orders for officers. The system then generates Blockchain Records (permanent proof), Analytics Reports (performance data), Audit Logs (action history), and Notifications (status updates). Every complaint is verified by neighbors, assigned to the right department, tracked transparently, and stored securely.



PAGE DESIGN

Mobile App Design



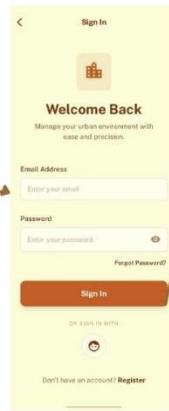
Splash screen



Authentication page



Registration page



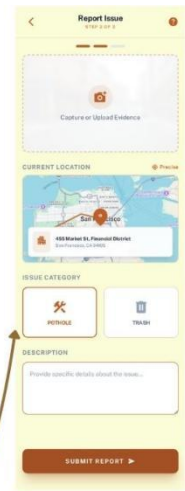
Sign In page



Home page



User Profile



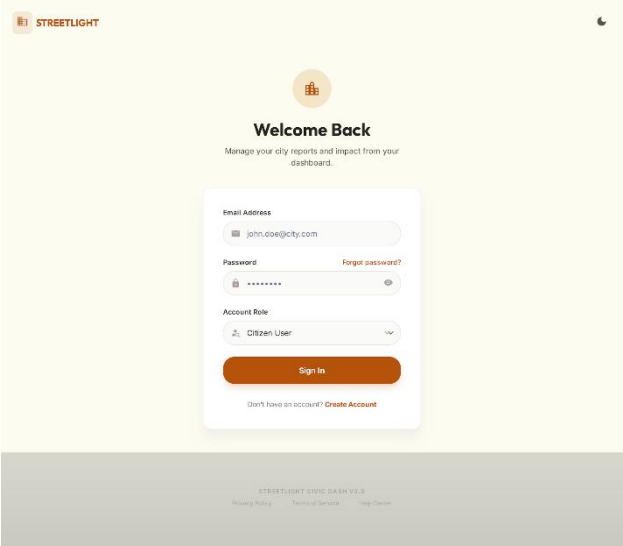
Report



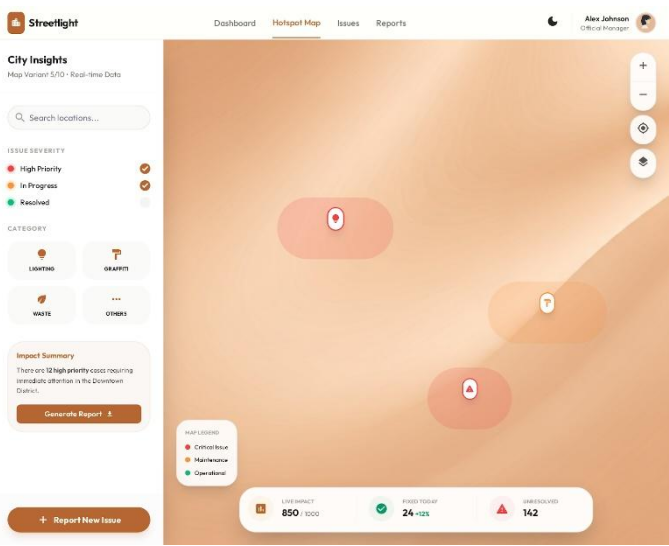
Explore

Web App Design

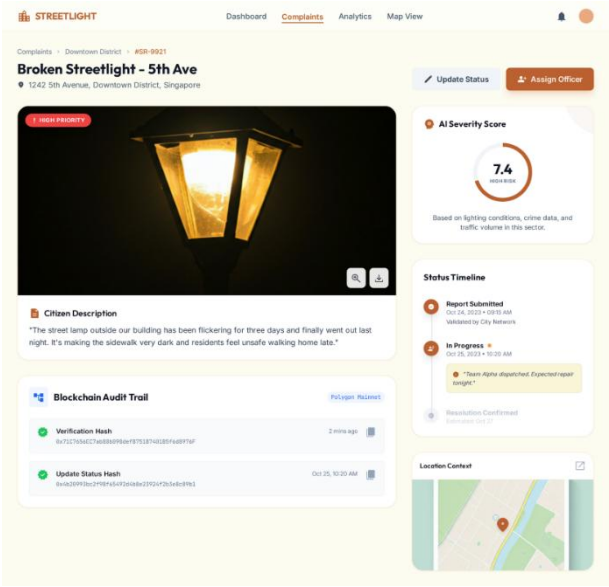
Sig-In page



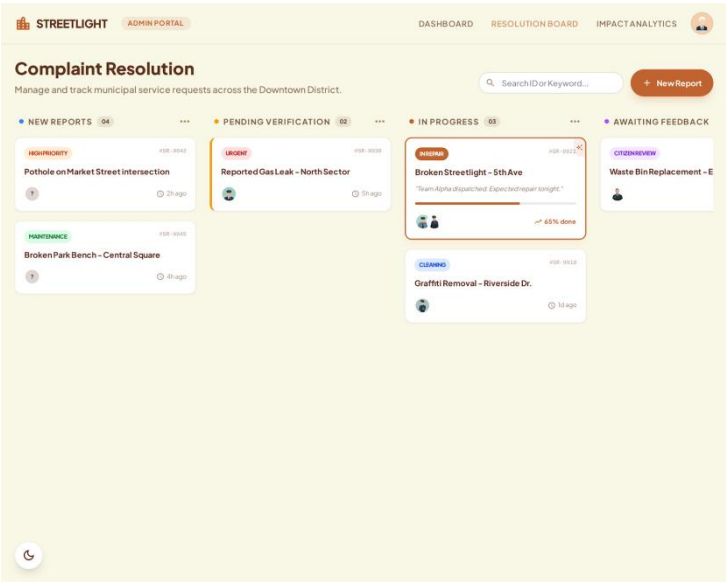
View Analytics Dashboard



Update Complaint Status

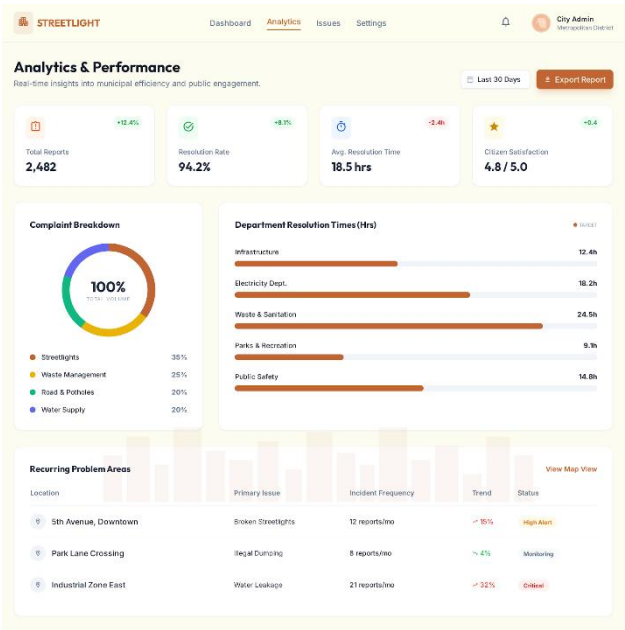


Assign Complaint to Officer



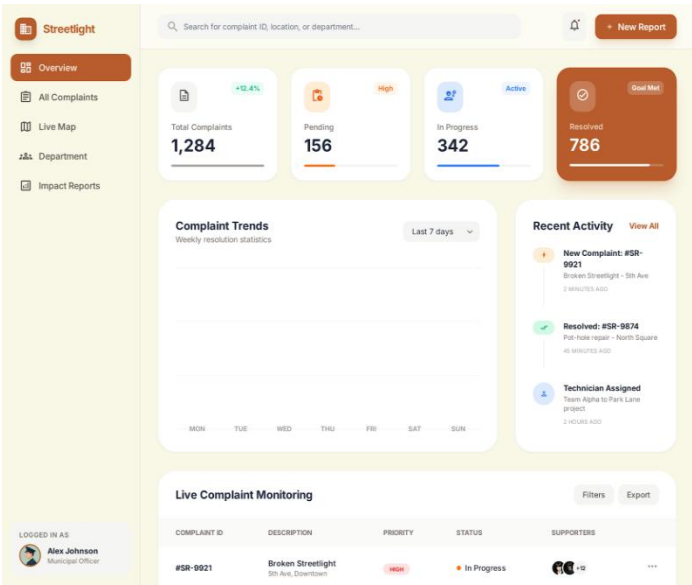
View Analytics Dashboard

(Performance & Insights)



View Analytics Dashboard

(Municipal Overview)



CASE TOOLS

Category	Tool Name	Reason for Selection
UML Diagrams	Draw.io	Free tool with online and offline support; supports all UML diagram types; easy to use; allows export in PNG, SVG, and PDF formats
Project Management	Jira	Widely used in agile development; supports customizable boards, issue tracking, and sprint planning
Documentation	MS Word	Industry-standard documentation tool; supports professional formatting, tables, and diagrams; widely familiar
Repository Management	GitHub	Free for students; provides version control, CI/CD integration, private repositories, and issue tracking
UI/UX Design	Figma	Free for students; supports wireframing and prototyping; real-time collaboration; reusable component libraries